

6.1.2 Charge and field

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i>	
(i) uniform electric field $E = V/d$	<i>M2.3</i>
(ii) the electric field of a charged object, and the force on a charge in an electric field; inverse square law for point charge	Spherically symmetrical charged conductor is equivalent to a point charge at its centre
(iii) electrical potential energy and electric potential due to a point charge; $1/r$ relationship	
(iv) evidence for discreteness of charge on electron	Such as the Millikan oil drop experiment HSW2, 7, 11
(v) the force on a moving charged particle due to a uniform magnetic field	
(vi) similarities and differences between electric and gravitational fields.	HSW2, 8

(b) *Make appropriate use of:*

- (i)** the terms: charge, electric field, electric potential, equipotential surface, electronvolt

by sketching and interpreting:

- (ii)** graphs showing electric potential as area under a graph of electric field versus distance, graphs showing changes in electric potential energy as area under a graph of electric force versus distance between two distance values. M3.8

- (iii)** graphs showing force as related to the tangent of a graph of electric potential energy versus distance, graphs showing field strength as related to the tangent of a graph of electric potential versus distance M3.8

- (iv)** diagrams of electric fields and the corresponding equipotential surfaces.

(c) *Make calculations and estimates involving:*

- (i)** for radial components

$$F_{electric} = \frac{kqQ}{r^2}, \quad E_{electric} = \frac{F_{electric}}{q}$$
$$= \frac{kQ}{r^2} \left[k = \frac{1}{4\pi\epsilon_0} \right]$$

- (ii)** $E_{electric} = -\frac{dV_{electric}}{dr},$

$$E_{electric} = \frac{V}{d} \quad (\text{for a uniform field})$$

- (iii)** electrical potential energy = $\frac{kQq}{r},$

$$V_{electric} = \frac{kQ}{r}$$

- (iv)** $F = qvB.$ M2.3